

Trees Still Beat Transformers for Day-Ahead Electricity Price Forecasting

M. Q. Ansari & M. Q. Ansari • PJM & ERCOT • 2019–2025 • 18 Models • Applied Energy

DATA

PJM + ERCOT (2 US markets)
61K hours × 50 features
4 volatility regimes

18 CONFIGURATIONS

3 Trees · 6 DL · 2 Foundation
Chronos-Bolt v1 & v2+Cov
Rolling retraining (24 mo)

TRIPLE EVALUATION

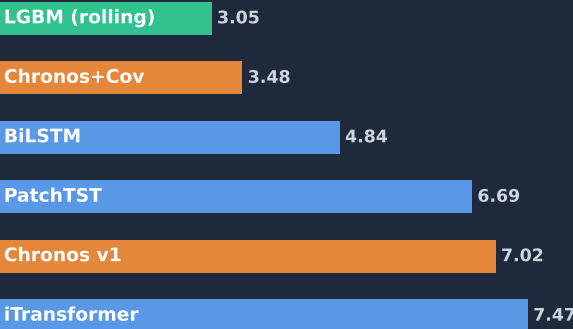
Point: MAE, RMSE, SMAPE
Probabilistic: CQR, QRF, MC-DO
Economic: PnL, Sharpe, Win%

STATISTICAL RIGOR

DM test + HLN correction
Benjamini–Hochberg FDR
24-window robustness

KEY FINDINGS

MAE (\$/MWh) — PJM



3.05

MAE \$/MWh (PJM)

Covariates > Scale

Chronos Architecture Upgrade

v1 → v2: −2.3%

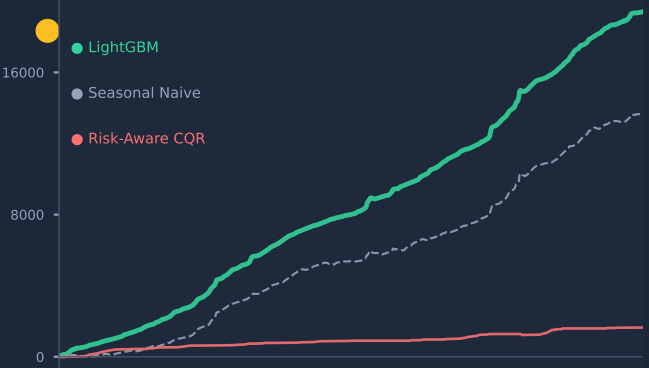
Adding Covariates

v2 → v2+Cov: −49%



6.85 → 3.48 \$/MWh

Cumulative PnL (PJM)

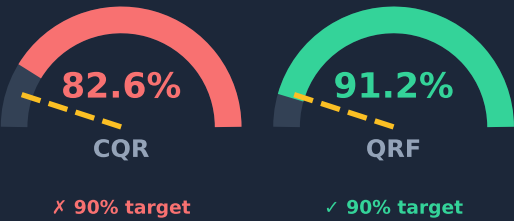


\$19,417

PnL • Sharpe 15.75 • 97.5% Win

Conformal Failure

Prediction Interval Coverage (PJM)



× 90% target

✓ 90% target

Exchangeability violated

Rolling LightGBM: MAE = 3.05 \$/MWh (PJM) | 2.58 \$/MWh (ERCOT)

Outperforming 17 alternatives including Chronos-2, PatchTST, iTransformer & TFT | $p < 0.01$ (DM + BH)

24.1%

Retraining gain

49%

Covariate gain

\$19,417

Trading PnL

91.2%

QRF coverage